

Problem Statement

Scilab Quantum Computing Visualizer (SQCV)

An Interactive GUI-Based Quantum Circuit Simulator in Native Scilab

Problem Being Addressed

Quantum computing is fundamentally counter-intuitive. Concepts such as superposition, entanglement, and phase kickback are described by high-dimensional complex vector spaces that are difficult to grasp from equations alone. Effective learning and rapid prototyping therefore demand interactive visualization tools that render abstract state vectors, unitary transformations, and measurement statistics into tangible graphical form.

Existing quantum simulation frameworks are predominantly built on Python-based ecosystems (e.g., Qiskit, Cirq) or web platforms, which introduces external dependencies, steep setup overhead, and a disconnect for users whose primary scientific computing environment is Scilab. Within the Scilab ecosystem, there is no native, end-to-end interactive tool that allows users to *construct* arbitrary quantum circuits, *simulate* them step-by-step, and *visualize* the evolving quantum state—including state-vector amplitudes, measurement probability histograms, and Bloch-sphere projections—without leaving the Scilab console. This gap limits the ability of students, educators, and researchers to explore quantum algorithms directly inside a familiar, open-source numerical computing platform.

Objective of the Application

The objective is to develop the **Scilab Quantum Computing Visualizer (SQCV)**: a self-contained, modular GUI application implemented entirely in Scilab that bridges quantum computational mathematics with real-time visual intuition.

The application must provide:

- **Interactive Circuit Designer:** A graphical canvas to place single-qubit gates (Hadamard, Pauli $X/Y/Z$, phase S/T , parametric rotations $R_X/R_Y/R_Z$), multi-qubit controlled gates (CNOT, CZ, Toffoli, Fredkin), and SWAP operations on up to 8 qubits with automatic wire-based layout.
- **Native Simulation Engine:** A pure-Scilab linear-algebra backend that evolves N -qubit state vectors using Kronecker-product gate expansion, controlled-projector operators, and density-matrix partial traces—with zero external toolbox dependencies.
- **Step-by-Step Execution:** A gate-by-gate stepper that highlights the active operation, logs the corresponding matrix transformation to the console, and updates all visualizations incrementally.
- **Integrated Visualization Panels:** (i) a state-vector inspector tabulating complex amplitudes, probabilities, and phases for all 2^N basis states; (ii) a probability bar chart that toggles between theoretical distributions and empirical 1000-shot Monte-Carlo histograms; and (iii) a 3D Bloch-sphere rendering of the reduced density matrix for any selected qubit.
- **Built-in Algorithm Demos:** One-click loading and execution of canonical protocols—Bell-state generation, Grover's search, Deutsch-Jozsa oracle evaluation, and quantum teleportation—to demonstrate entanglement, amplitude amplification, and state collapse in a pedagogical context.

All functionality must be accessible through a single-line launch command (`SQCV_start()`) and run on standard Scilab 6.x installations across Windows, Linux, and macOS.

Intended Users and Use Cases

- **Undergraduate and Graduate Students** in physics, computer science, and engineering programs who need an intuitive, click-and-see environment to build mental models of qubit behavior before advancing to hardware-specific SDKs.
- **Educators and Teaching Assistants** designing laboratory modules or classroom demonstrations on quantum information theory, requiring a stable, dependency-free simulator that runs on institutional Scilab installations without administrative privileges or internet access.
- **Researchers and Hackathon Participants** who use Scilab for control-system or signal-processing workflows and wish to prototype hybrid classical-quantum algorithms or verify gate decompositions rapidly within their existing toolchain.

By eliminating external dependencies and embedding the full simulation-visualization pipeline inside Scilab, SQCV lowers the barrier to quantum literacy and provides a reproducible, platform-agnostic educational instrument.